Tendering and measurement

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Topics to cover this week

1. From Procurement to Tender Package
2. Scope and Specifications
3. Measurement and Pricing
4. Tenderer Requirements and Submission
5. Tender Evaluation



What makes a project tenderable?

Procurement strategy: how the project will be bought
Tendering document: define the project well enough for contractors to compete for

- Defined scope: what is included and excluded?
- Specified requirements: what performance or standard is required?
- Pricable structure: what exactly should bidders price?
- Comparable submissions: what must every bidder return?
- Evaluable criteria: how will the client distinguish one bid from another?

e.g.
Improve the sustainability of the main building.
Replace existing corridor lighting on level 2 with LED and occupancy controls, while maintaining specified illuminance levels.



Tender package components

Information exchange designed to produce comparable offers

Client issues	Tenderer returns
Instructions to tenderers	Completed tender form
Scope of works	Technical proposal/method
Specifications and drawings	Proposed materials/equipment
Pricing document/BOQ/schedule	Completed pricing document
Program requirements/key dates	Proposed program
Qualification requirements	Experience, licenses, key personnel
Evaluation criteria	Information needed for evaluation



Internal consistency

Scope → **Specification** → **Pricing** → **Submission** → **Evaluation**

Scope requires an automated lighting control → Pricing schedule has no item for controls → Bidder prices differ in what they assume → Tenders are no longer directly comparable.

University requires all noisy work to be carried out outside teaching hours → the tendering document provides no item for night/weekend working → Bidder was awarded with a low price based on normal program → Over budget/deadline and dispute during execution.



Project objective ≠ scope of works

Project objective:

- Describes why the project is being undertaken
- States the intended outcome

Scope of Works:

- Defines what work the contractor is responsible for
- Establishes the project boundary

Example:
Objective: Reduce annual cooling energy consumption by 20% in the Composite building.
Scope: Design, supply, install, test and commission upgrades to the AHUs, controls, and associated sensors serving Floors 1-6, excluding chillers and facade works.

Is detailed scoping still necessary if a D&B approach is adopted?



Defining the project boundary

A good scope should make clear:

- **Location**: where are the works?
- **Extent**: which systems or assets are included?
- **Activities**: removal, supply, installation, modification, testing, etc.
- **Interfaces**: where does this work connect with existing systems or other parties?
- **Responsibilities**: what is provided by the contractor, client, or others?
- **Exclusions**: what is explicitly outside the contract?

Allow two competent bidders to form a structurally similar understanding of what work is required.



Gaps usually occur at interfaces

Many scope disputes arise from interfaces, not from the main work itself.

Included in contractor scope	By client / others	Excluded
Examine existing AHU and duct network	Provide access to plant room	Remove unused equipment
Supply and install new sensors, controllers and associated wiring	Approve shutdown periods and access arrangements	Repair of pre-existing faults in existing HVAC equipment
Test and commission the revised control system	Provide required BMS access and credentials	Recommissioning of unaffected air-side systems



How much detail is enough?

Upgrade the ventilation system to improve indoor air quality.

↓

Replace existing exhaust fans and install additional ventilation controls in Rooms 1-5.

↓

Remove four existing exhaust fans, supply and install replacement units serving Rooms 1-5, integrate the operation with the existing control system, balance airflow, test performance, and submit commissioning records.

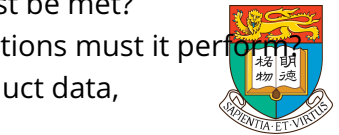
What information is still missing before this can be priced confidently?



From scope to specification

	Scope	Specification
Main question	What must be done?	What standard must it meet?
Example	Replace 100 lights	Minimum efficiency, illuminance, and lifetime
Focus	Work boundary	Technical/performance requirement

- **Functional requirements**: what must the system be able to do?
- **Technical requirements**: capacities, materials, components, protocols, etc.
- **Compatibility requirements**: how must it interface with existing systems?
- **Standards and codes**: what regulations or standards must be met?
- **Environmental/operating conditions**: under what conditions must it perform?
- **Documentation requirements**: drawings, manuals, product data, commissioning records



How much should the client specify

Prescriptive specification: defines materials, products, dimensions or methods

- Client retains more control over the solution
- Easier to compare compliance
- May limit innovation
- More design responsibility with client

Performance specification: defines required outcomes

- Contractor has more freedom in how to achieve them
- Encourages alternative solutions
- More design responsibility with contractor
- *Requires measurable performance criteria*

Specification style should be consistent with procurement route and allocation of design responsibility.



Make requirements measurable

A requirement is useful only if compliance can be checked.

Weak requirement	**More measurable requirement**
Use energy-efficient lighting	Luminaire efficacy $\geq$ 120 lm/W
Improve indoor comfort	Maintain operative temperature within specified range
Minimize construction waste	Divert $\geq$ 80% of construction waste from landfill
Use high-quality equipment	Equipment to comply with specified standard and warranty
Minimize disruption	No noisy works during scheduled teaching hours

Requirement $\rightarrow$ Verification method $\rightarrow$ Acceptance



From scope to price

Given the defined work, how does the client get comparable prices from contractors?

Measurement and pricing documents help to:

- Translate the scope into priceable work items
- Define a common pricing basis for all bidders
- Make tender prices transparent and comparable
- Provide a basis for payment and valuation of changes after award

Example: upgrade AHU control → sensors, controllers, wiring, BMS integration, testing and commissioning



Break work down

Translates the scope into defined work items with quantities and units.

e.g. upgrade AHU control

Item	Description	Quantity	Unit
1	CO_2 sensors	12	no.
2	Temperature sensors	18	no.
3	Control panels	3	no.
4	Control cabling	250	m
5	BMS integration	1	item
6	Testing and commissioning	1	item

Measurement is not just about quantities. It creates a common commercial structure for bidding and subsequent valuation.

When should items be separated?

More detail does not automatically mean better measurement. Separate items when they matter commercially.

e.g. upgrade AHU control

- Temperature sensor mounting bracket
- Temperature sensor label
- Temperature sensor fixing screws
- Cable clips
- Cable ties

Would separating this item improve pricing clarity, tender comparison, payment, or valuation of changes?



Ways to structure the tender price

Pricing document	How bidder prices	Suitable when
Bill of Quantities	Rate × client-provided quantities	Work can be measured reasonably clearly
Schedule of Rates	Bidder provides unit rates	Quantities are uncertain or work may be instructed as needed
Activity Schedule	Price assigned to defined activities/work packages	Work is better structured by activities than detailed quantities
Lump-sum breakdown	Prices for major components of a fixed scope	Scope is well defined and detailed measurement is unnecessary



Who provides what?

Client prepares: item descriptions, pricing format, quantities, units, instructions on what prices include

Tenderer completes: unit rates, item amounts, lump-sum prices, required price breakdown, tender total

Item	Description	Quantity	Unit	Rate	Amount
1	CO_2 sensors	12	no.	-	-
2	Temperature sensors	18	no.	-	-
3	Control panels	3	no.	-	-
4	Control cabling	250	m	-	-
5	BMS integration	1	item	-	-
6	Testing and commissioning	1	item	-	-



Double check consistency

Scope → Specification → Pricing must remain internally consistent.

Scope: Supply, install, integrate, test and commission new HVAC controls.
Pricing schedule: Lists only sensors and controllers as equipment items, with no installation, BMS integration, or commissioning items, and no instruction that these costs must be included in the equipment rates.

Bidder A: includes installation and commissioning within unit rates.
Bidder B: adds separate allowances.
Bidder C: omits manpower entirely.
→unreliable comparison



Pricing instructions

Typical tender instructions may state that prices include:

- Labor
- Materials and equipment
- Plant and tools
- Transport and delivery
- Installation
- Temporary works
- Testing and commissioning
- Waste disposal
- Overheads and profit



What's needed from the tenderer?

Qualification information + Technical/management proposal + Commercial offer

e.g. upgrade AHU control

- **Qualification**: relevant BMS/HVAC experience, required registration
- **Technical**: proposed controls solution, equipment, integration approach
- **Management**: program, method statement, project team
- **Commercial**: completed pricing schedule and tender sum



Qualification vs Tender proposal

Can they do it? vs How will they do it?

	Qualification	**Tender proposal**
Personnel	Does the contractor have qualified engineers and technicians?	Which personnel will be assigned, and what are their project roles?
Technical capability	Can the contractor undertake BMS integration and controls work?	What control architecture, equipment and integration approach are proposed?
Safety	Does the contractor have an acceptable safety management system and record?	How will safety be managed during works in an occupied campus building?



Typical tender submission requirements

Qualification information

Relevant experience, licenses/registrations, financial standing, safety capability/record, key personnel/capability, key subcontractors/supplier

Technical/management proposal

Technical proposal, equipment/materials, work program, project team and resources, testing/commissioning approach

Commercial offer

Completed pricing document, tender sum, price breakdown, pricing basis, assumptions/conditions

Mandatory requirements vs information for evaluation

Determine whether a tender is acceptable → compare acceptable offers

	Purpose	**Example**
Compliance check	Confirm minimum (technical) requirements are satisfied	Signed forms, required documents, system compatibility
Technical evaluation	Compare the quality of acceptable bids	Methodology, program, team experience, proposed solution
Commercial assessment	Compare price and commercial basis	Tender sum, completed pricing schedule, pricing assumptions

Difference between compliance and qualification?



Instructions to tenderer

Typical instructions should state:

- Tender closing date and submission method
- Required tender documents
- Format and organization of submission
- Tender validity period
- Procedure for tender queries
- Site visit/briefing arrangements, if applicable
- Rules for alternative proposals, if permitted
- Required signatures and declarations



Ask for useful information

Why are we asking bidders to prepare it?

Examples of poor queries:

- Require a detailed sustainability plan, while sustainability is neither a compliance requirement nor an evaluation criterion
- Evaluation gives 10% to program quality, but a program was not explicitly required
- “Provide evidence of relevant experience.”
- “Submit full technical details of the proposed system.”



Overall tender evaluation process

Tender A	Tender B
HK$1.00M	HK$1.08M
Meets minimum requirements	Stronger technical solution
10-week programme	14-week programme
Limited occupied-building experience	Extensive occupied-building experience

Client priorities → Evaluation criteria → Weightings → Tender scores → Award decision



What makes good evaluation criteria?

- **Relevant**: linked to an actual project objective, constraint, or delivery need
- **Discriminating**: capable of revealing meaningful differences between bids
- **Evidence-based**: supported by information that bidders are explicitly required to submit
- **Specific enough to score consistently**: different evaluators should interpret it similarly
- **Non-overlapping**: avoid scoring the same attribute twice
- **Proportionate**: the effort required from bidders should match the importance of the criterion



Evidence-based evaluation

Evaluate a criterion consistently based on required evidence that supports it.

Evaluation criterion	Typical supporting evidence
Technical solution	Technical proposal, equipment schedule, system architecture, datasheets
Relevant experience	Similar completed projects, project value, scope, completion date, client reference
ESG/performance	Energy-performance calculations, proposed efficiency improvements, monitoring plan

Example: Criterion: "Experience - 10%"
Submission requirement: "Provide company profile."
↓
Submit up to three comparable HVAC controls retrofit projects, including scope, contract value, completion date, project role, and client reference.



Weighting and scoring

Scoring scale example: 0 - Unacceptable; 1 - Major weaknesses; 2 - Partially meets requirement; 3 - Meets requirement; 4 - Exceeds requirement; 5 - Excellent/clearly exceeds requirement.

Not all criteria matter equally

Criterion	Weight
Price	45%
Technical solution	25%
Methodology/program	15%
Relevant experience/team	10%
Performance/quality	5%

Weighted score = raw score × criterion weight



Price evaluation

Before scoring price, check: completeness, pricing assumptions/exclusions, unbalanced pricing, abnormally low price, etc.

Establish whether prices are comparable, potentially adjust with predefined rules:

	Tender A	**Tender B**
Submitted tender sum	HK$1.00M	HK$0.94M
Commissioning omitted	included	+HK$0.08M
Comparable evaluated price	HK$1.00M	HK$1.02M

Quantitative scoring, example:

$$\textit{Price score} = \frac{\textit{lowest evaluated price}}{\textit{bidder's price}} \times \textit{maximum score}$$



Common evaluation mistakes

A good evaluation system is transparent, project-specific, and applied consistently to comparable tenders:

- **Changing criteria or weights after seeing the bids**: The program was originally worth 10%, but the evaluators later decided it should matter much more.
- **Double counting the same attribute**: Relevant experience is scored under both “Contractor Capability” and “Technical Quality.”
- **Weightings do not reflect project priorities**: For an occupied teaching building, program and disruption management receive only 5%, while generic ESG receives 20%.
- **Too vague scoring descriptors**: ”1 = Poor, 3 = Good, 5 = Excellent”

